

PE-EC802A — Mixed Signal Design

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Module 2 Exam Notes

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1. Switched-Capacitor (SC) Circuits & Resistor Equivalence

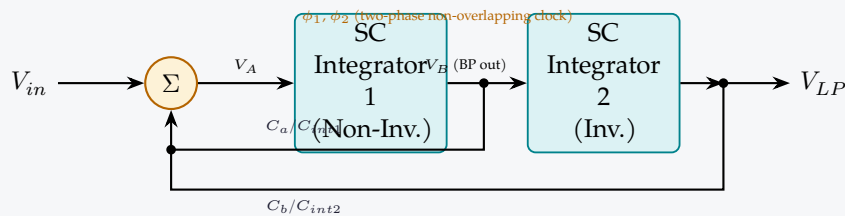
Switched-Capacitor Circuit Concept

A **Switched-Capacitor (SC) circuit** is a discrete-time signal processing circuit that implements resistors by rapidly switching charge back and forth between capacitors using non-overlapping clock phases. SC circuits are highly preferred in CMOS technology because precise on-chip resistors are difficult to fabricate, whereas capacitor ratios can be matched with extremely high accuracy (up to 0.1%).

1.1 Switched-Capacitor Filter Architectures

A practical SC biquad (second-order section) is built using two SC integrators in a feedback loop. The **Fleischer-Laker biquad** is the standard SC biquad implementing LP, HP, and BP outputs simultaneously.

Fleischer-Laker SC Biquad Block Structure



How the Fleischer-Laker Biquad Sets Frequency and Q:

- Pole frequency: $\omega_0 \propto \frac{f_{clk} \cdot C_{sampling}}{C_{integrating}}$ — set purely by capacitor ratios and clock frequency.
- Quality factor Q : set by the feedback capacitor ratio. Changing C_a and C_b independently controls ω_0 and Q without interaction.

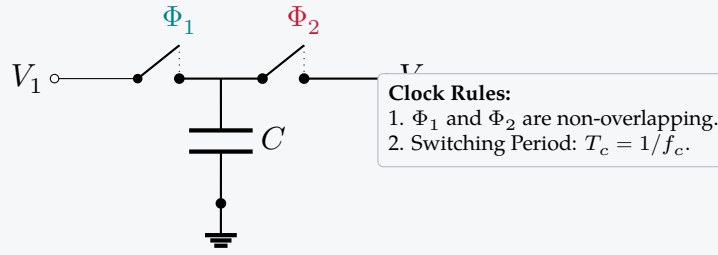
1.2 Switched-Capacitor Filter Applications

- **Audio CODEC:** Anti-aliasing and reconstruction filters in 16–24-bit audio codecs (e.g., CD player, smartphone DAC) where precise $\pm 0.01\%$ corner frequency tolerance is required.
- **$\Delta\Sigma$ ADC Front-End:** SC integrators inside $\Delta\Sigma$ modulators perform the summation and integration operations that shape quantization noise to high frequencies.
- **DTMF (Dual-Tone Multi-Frequency) Decoding:** Narrow-band bandpass SC filters detect individual telephone dial tones.
- **Biomedical Instrumentation:** Low-frequency, high-accuracy SC filters for ECG/EEG signal conditioning, where process-insensitive frequency accuracy is essential.

1.3 Derivation of SC Resistor Equivalence

Consider a capacitor C connected between two voltage nodes V_1 and V_2 via two single-pole single-throw switches controlled by non-overlapping clock phases Φ_1 and Φ_2 with a clock frequency $f_c = 1/T_c$.

Switched-Capacitor Resistor Equivalence Circuit



Step-by-step Derivation:

1. During phase Φ_1 , Switch 1 is closed and Switch 2 is open. The capacitor C charges to voltage V_1 . The charge stored on the capacitor is:

$$Q_1 = CV_1$$

2. During phase Φ_2 , Switch 1 opens and Switch 2 closes. The capacitor C is connected to V_2 , charging or discharging to V_2 . The charge remaining on the capacitor is:

$$Q_2 = CV_2$$

3. The net charge transferred from node V_1 to node V_2 in one clock period T_c is:

$$\Delta Q = Q_1 - Q_2 = C(V_1 - V_2)$$

4. The average current I_{avg} flowing from node V_1 to node V_2 is the charge transferred per unit time:

$$I_{avg} = \frac{\Delta Q}{T_c} = \frac{C(V_1 - V_2)}{T_c} = f_c C(V_1 - V_2)$$

5. Using Ohm's Law ($I = \frac{V_1 - V_2}{R_{eq}}$), we find the equivalent resistance R_{eq} of the switched-capacitor stage:

SC Equivalent Resistance

$$R_{eq} = \frac{V_1 - V_2}{I_{avg}} = \frac{1}{f_c C}$$

where:

- R_{eq} = Equivalent resistance (Ω)
- f_c = Clock switching frequency (Hz)
- C = Capacitance (F)

1.4 Significance of SC Circuits in IC Design

In integrated circuits, fabricating large resistors (e.g., 100 k Ω) is highly impractical because:

- **Area:** Polysilicon/diffusion resistors require massive silicon area.
- **Accuracy:** Absolute tolerances of resistors on silicon are poor ($\pm 20\%$) and drift significantly with temperature.

- **Tuning:** RC time constants ($\tau = RC$) vary by up to 40% due to process variations.

In contrast, in SC circuits, the equivalent time constant of an integrator is:

$$\tau = R_{eq}C_2 = \left(\frac{1}{f_c C_1} \right) C_2 = \frac{1}{f_c} \left(\frac{C_2}{C_1} \right)$$

Since f_c is derived from a highly stable crystal oscillator and $\frac{C_2}{C_1}$ is a ratio of two matched capacitors (which matches perfectly within 0.1% on layout), the time constant τ is extremely accurate and temperature-independent.

2. SC Non-Idealities & Mitigation Strategies

In real CMOS processes, MOS transistors act as non-ideal switches. The key non-idealities are:

2.1 1. Charge Injection

When a MOS switch is turned ON, a channel charge Q_{ch} is accumulated in the inversion layer:

$$Q_{ch} = WLC_{ox}(V_{GS} - V_{TH}) = WLC_{ox}(V_{CLK} - V_{in} - V_{TH})$$

When the switch is turned OFF (gate voltage drops), this channel charge is released and flows out through the source and drain terminals. The charge injected into the sampling capacitor C shifts the sampled voltage, introducing an offset and non-linear distortion (since V_{TH} depends non-linearly on V_{in} via the body effect).

2.2 2. Clock Feedthrough

The overlap capacitance C_{ov} between the gate and source/drain terminals forms a capacitive voltage divider with the sampling capacitor. The clock transition at the gate is directly coupled to the sampling capacitor:

$$\Delta V = V_{CLK} \left(\frac{C_{ov}}{C_{ov} + C} \right)$$

This introduces a constant offset voltage to the sampled signal.

2.3 3. thermal Noise ($k_B T/C$ Noise)

An ON transistor has a finite channel resistance R_{on} . This resistance generates thermal noise with a power spectral density of $4k_B T R_{on}$. When the switch is opened, this noise is sampled onto the capacitor. The total noise power integrated over the loop bandwidth is:

Thermal Noise Limit (kT/C)

$$v_n^2 = \int_0^\infty \frac{4k_B T R_{on}}{1 + (2\pi f R_{on} C)^2} df = \frac{k_B T}{C}$$

where k_B is the Boltzmann constant (1.38×10^{-23} J/K) and T is temperature in Kelvin.

This noise sets a fundamental limit on the dynamic range (SNDR) of switched-capacitor circuits. To achieve higher resolution, C must be made larger, which increases power consumption and area.

Summary of Switch Mitigation Techniques

Non-Ideality	Physical Mechanism	Mitigation Strategy
Charge Injection	Channel charge flows into C upon turn-off	Dummy switches, differential signaling, bottom-plate sampling
Clock Feedthrough	Overlap capacitance C_{ov} couples clock edges	CMOS transmission gates, differential architectures
Thermal Noise	Thermal noise of R_{on} is sampled onto C	Increase capacitance value C ($v_n \propto 1/\sqrt{C}$)

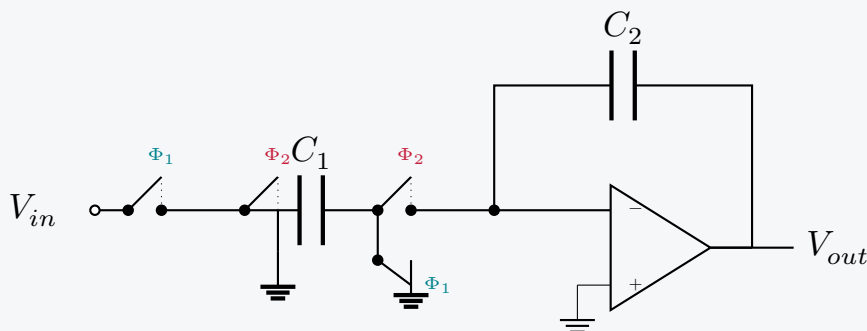
3. Stray-Insensitive Integrators

Early SC designs were sensitive to parasitic capacitances between node terminals and substrate (ground), causing gain errors. Modern SC systems use **stray-insensitive** configurations.

3.1 Stray-Insensitive Non-Inverting Integrator

The stray-insensitive non-inverting integrator uses 4 switches controlled by non-overlapping clocks Φ_1 and Φ_2 .

Stray-Insensitive Non-Inverting SC Integrator Circuit



Analysis of Stray Insensitivity:

- **Node A (left plate of C_1):** Connected to either V_{in} (low impedance source) during Φ_1 or to ground during Φ_2 . Any parasitic capacitor at this node is either charged by V_{in} or fully discharged to ground, never affecting the charge transferred to the integrating capacitor C_2 .
- **Node B (right plate of C_1):** Connected to virtual ground (op-amp inverting terminal) during Φ_2 or to physical ground during Φ_1 . The potential is always at ground (virtual or physical), so the voltage across its parasitics is always 0 V. Thus, no charge is stored or injected by the parasitic capacitance.

3.2 Charge Equations and Transfer Function

Let us write the discrete-time charge conservation equation at the virtual ground node:

1. During $\Phi_1(n-1)$, C_1 charges to $V_{in}(n-1)$ (left terminal to V_{in} , right terminal to ground).

$$q_{C_1}(n-1) = C_1 V_{in}(n-1)$$

2. During $\Phi_2(n)$, C_1 is connected to ground on the left and virtual ground on the right. The capacitor is fully discharged, and its charge flows entirely into C_2 . The feedback capacitor C_2 accumulates this charge:

$$C_2 V_{out}(n) = C_2 V_{out}(n-1) + C_1 V_{in}(n-1)$$

3. Taking the Z-transform of this difference equation:

$$C_2 Y(z) = C_2 z^{-1} Y(z) + C_1 z^{-1} X(z)$$

$$Y(z)(1 - z^{-1}) = \frac{C_1}{C_2} z^{-1} X(z)$$

4. The transfer function $H(z) = Y(z)/X(z)$ is:

Non-Inverting SC Integrator Transfer Function

$$H(z) = \frac{V_{out}(z)}{V_{in}(z)} = \frac{C_1}{C_2} \frac{z^{-1}}{1 - z^{-1}}$$

The term z^{-1} in the numerator indicates a delay of one clock cycle, making this a **delaying (non-inverting) integrator**.

If we modify the switch network such that C_1 is connected to V_{in} and virtual ground during the same clock phase, we obtain an **inverting (non-delaying) integrator** with transfer function:

$$H(z) = -\frac{C_1}{C_2} \frac{1}{1 - z^{-1}}$$

4. First-Order & Second-Order SC Filters (Biquads)

4.1 Fleischer-Laker Biquad Topology

The Fleischer-Laker biquad is a standard stray-insensitive second-order SC filter stage. It consists of two op-amps and a network of switched and unswitched capacitors. It can implement low-pass, high-pass, band-pass, and notch transfer functions by selecting appropriate capacitor ratios.

The general Z-domain transfer function of a second-order SC biquad is given by:

$$H(z) = \frac{N(z)}{D(z)} = \frac{\alpha_0 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}{1 + \beta_1 z^{-1} + \beta_2 z^{-2}}$$

By mapping s-plane specifications (e.g. pole frequency ω_0 and quality factor Q) through the Bilinear Transformation, the discrete coefficients α_i and β_i are determined, allowing direct calculation of the required capacitor ratios.

Quick Revision — Module 2

Switched-Capacitor Filters | PE-EC802A

Key Formulas at a Glance

Equivalent Resistance	$R_{eq} = \frac{1}{f_c C}$
Integrator Time Constant	$\tau = \frac{1}{f_c} \left(\frac{C_2}{C_1} \right)$
Thermal Noise Power	$v_n^2 = \frac{k_B T}{C}$
Non-Inverting SC Integrator	$H(z) = \frac{C_1}{C_2} \frac{z^{-1}}{1-z^{-1}}$
Inverting SC Integrator	$H(z) = -\frac{C_1}{C_2} \frac{1}{1-z^{-1}}$
Channel Charge	$Q_{ch} = WLC_{ox}(V_{CLK} - V_{in} - V_{TH})$
Clock Feedthrough Offset	$\Delta V = V_{CLK} \left(\frac{C_{ov}}{C_{ov}+C} \right)$

One-Line Definitions

- **SC Resistor Equivalence:** A capacitor switched at high rate acts as a resistor of value $1/(f_c C)$.
- **Non-overlapping Clocks:** Clock signals Φ_1 and Φ_2 that are never high at the same time to prevent direct shorting.
- **Charge Injection:** Channel charge of a switching MOS transistor entering the sampling capacitor upon turn-off.
- **Clock Feedthrough:** Direct capacitive coupling of the clock transition to the node via overlap capacitance.
- **Stray-Insensitive Integrator:** SC integrator topology immune to parasitics of capacitors to substrate/ground.
- **kT/C Noise:** The fundamental thermodynamic limit on sampled voltage accuracy, equal to $k_B T/C$.
- **Stray-Insensitive SC Integrator:** An SC integrator topology where parasitic capacitances at the virtual ground node of the op-amp are shielded and do not contribute to the transfer function.

Continuous-Time vs. Switched-Capacitor Filter Comparison

Property	Active RC Filter	Switched-Capacitor Filter
ω_c accuracy	Depends on absolute R, C ($\pm 20\%$)	Set by C ratio $\times f_{clk}$ ($< 0.1\%$)
Tuning	Manual trim required	Reprogram f_{clk}
Frequency range	DC to > 1 GHz	DC to $\sim f_{clk}/2$ (audio/kHz range)
Clock noise	Not applicable	Clock feedthrough adds noise floor
Silicon area	Needs large R, C	Compact (capacitors only)
CMOS compatible	Requires resistors	Fully CMOS-compatible

Frequently Asked / Viva Questions

Q1. Derive the equivalent resistance of a switched capacitor.

A capacitor C charged to V_1 at phase Φ_1 holds charge $Q_1 = CV_1$. At phase Φ_2 , it discharges to V_2 , holding charge $Q_2 = CV_2$. The net charge transferred per cycle is $\Delta Q = C(V_1 - V_2)$. The average current is $I_{avg} = \Delta Q/T_c = f_c C(V_1 - V_2)$. Since $R_{eq} = (V_1 - V_2)/I_{avg}$, we get $R_{eq} = 1/(f_c C)$.

Q2. Why are switched-capacitor circuits preferred over continuous-time active RC circuits on-chip?

Precise resistors are hard to fabricate in CMOS (requiring huge area and having $\pm 20\%$ variations). SC circuits replace resistors with capacitors. The time constant depends on the ratio of two matched capacitors, which matches within 0.1% , providing highly precise and temperature-stable performance.

Q3. What is kT/C noise and what limits does it impose?

It is the thermal noise power sampled onto a capacitor from the switch channel resistance. It is independent of the switch resistance and equals $k_B T/C$. It sets a lower limit on signal resolution; higher dynamic range requires larger capacitors.

Q4. Explain the mechanism of charge injection in MOS switches.

When an MOS transistor is ON, it has charge in its inversion channel. When it turns OFF, this charge exits through the source and drain terminals, injecting charge into the sampling capacitor, shifting the voltage and causing non-linear distortion.

Q5. How does a dummy switch mitigate charge injection?

A dummy transistor (usually half the size of the main switch with source/drain shorted) is connected in series. It is driven by the complementary clock. When the main switch turns OFF and releases charge, the dummy switch turns ON and absorbs that charge, neutralizing the error.

Q6. What is bottom-plate sampling, and why is it stray-insensitive?

It is a switching technique where the switch connected to the virtual ground (bottom plate) is opened slightly before the input switch. Since the bottom plate

Frequently Asked / Viva Questions (contd.)

is at virtual ground, the charge injected is constant and independent of the input voltage, eliminating non-linear distortion.

Q7. Show the Z-domain transfer function of a non-inverting SC integrator.

It is $H(z) = \frac{C_1}{C_2} \frac{z^{-1}}{1-z^{-1}}$, which represents a delaying integrator.

Q8. Compare inverting and non-inverting SC integrator transfer functions.

The inverting integrator is non-delaying: $H(z) = -\frac{C_1}{C_2} \frac{1}{1-z^{-1}}$. The non-inverting integrator has a delay: $H(z) = \frac{C_1}{C_2} \frac{z^{-1}}{1-z^{-1}}$.

Q9. What is the Fleischer-Laker biquad?

It is a stray-insensitive, second-order switched-capacitor filter topology capable of implementing any generic second-order transfer function (LP, HP, BP, Notch) by choosing appropriate capacitor values.

Q10. What is clock feedthrough and how is it reduced?

Clock feedthrough is the coupling of the clock gate signal to the sampling capacitor via the MOS gate-overlap capacitances. It is reduced by using transmission gates (where PMOS and NMOS cancel each other's transitions) or differential architectures.